EE2015: Electric Circuits and Networks

Tutorial 5 Answer Key
(September 19, 2025)

1. $\mathbf{h} = \begin{bmatrix} 4\,\Omega & \frac{2}{3} \\ -\frac{2}{3} & \frac{1}{9}\,\mathrm{S} \end{bmatrix}$

2. $R_{\text{Th}} = 51.46\,\Omega$,
   $V_{\text{Th}} = -29.69$

3. $\mathbf{h} = \begin{bmatrix} 3.8\,\Omega & 0.4 \\ -3.6 & 0.2\,\mathrm{S} \end{bmatrix}$

4. $\mathbf{g} = \dfrac{1}{s^2 + 3s + 1} \begin{bmatrix} s+2 & -1 \\ 1 & s(s+2) \end{bmatrix}$

5. $\mathbf{h} = \begin{bmatrix} 85\,\Omega & 0.25 \\ 14.75 & 0.0725\,\mathrm{S} \end{bmatrix}$

   $\mathbf{g} = ?$

6. $\mathbf{t} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

7. $R_L = 8\,\Omega$, $3.125\,\mathrm{W}$